

Fundamental concepts and electrons in atoms

Dalton: 1766-1844

- > All elements composed of tiny particles called **atoms**
- > Atoms of same element are identical; atoms of different elements are different
- > Atoms of different elements can physically mix together or chemically combine to form compounds
- > Chemical reactions **cannot** change atoms of one type of element to another



Thomson: 1856-1940

- > discovered electrons in 1897
- > used a cathode ray tube
- > the ray produced was deflected by an electrical field (showed that atoms had particles with (-) charge)



Cathode Ray Tubes

- A cathode ray tube or CRT is a specialized vacuum tube in which images are produced when an electron beam strikes a phosphorescent surface.
- TVs, PCs, ATMs, video games, video cameras, and monitors all contain cathode-ray tubes.
- Displays millions of colors.

Rutherford: 1871-1937

- > Gold Foil Experiment
- > Discovered the nucleus



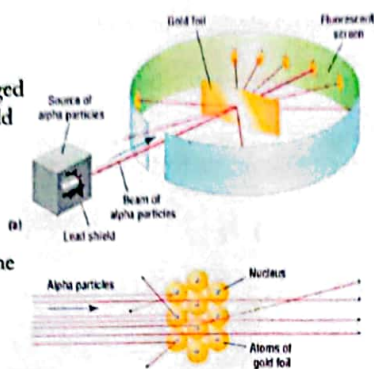
Rutherford's Gold Foil Experiment

Experiment

Shot positively charged alpha particles at gold foil

Results

- 1) Most particles passed through the foil
- 2) A few were deflected



Rutherford's Gold Foil Experiment

Conclusions

- 1) small, dense, positively charged core (nucleus)
- 2) the rest of the atom is empty space

The Bohr Model

- Niels Bohr studied the hydrogen atom because it was the most simplistic.
- Bohr proposed that an electron is found only in specific circular paths, or orbits, around the nucleus.
- Each possible electron orbit in Bohr's model has a fixed energy. The fixed energies an electron can have are called energy levels.



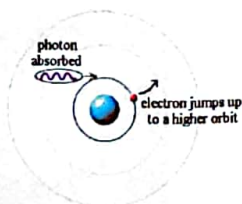
The Bohr Model

- The energy levels get closer together as you move farther from the nucleus.
- The energy levels also get higher in energy as you move farther from the nucleus.



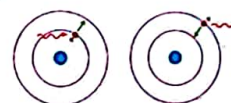
Electrons Jump

- Electrons can jump from one energy level to another.
- A quantum of energy is the amount of energy required to move an electron from one energy level to another.



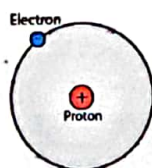
Electrons Jump

- An electron must gain energy to jump to a higher energy level.
- When an electron has jumped to a higher energy level, it is in an excited state.
- An electron must lose energy to fall to a lower energy level.
- When an electron is at the lowest energy level possible, it is at ground state.



Bohr's Model Restrictions

- Bohr's model accurately describes the movement of an electron in the hydrogen atom, but it cannot describe the movement of multi-electron atoms.



Quantum Mechanical Model

- The quantum mechanical model of the atom is based on the mathematical probability of the location of electrons using the Schrodinger equation.
- The quantum mechanical model stills has energy levels, but the exact path or orbit of the electron is unknown.



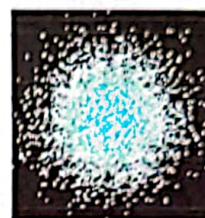
Quantum Mechanical Model

- Since the quantum mechanical model is based on the probability of finding an electron, then the orbitals are normally shaded with a fuzzy edge.
- An atomic orbital is a region of space in which there is a high probability of finding an electron.



Atomic Orbitals

- Within the border of an atomic orbital, there is a 90% chance of finding an electron.
- The darker the shading of the orbital, the higher the chance of finding an electron.

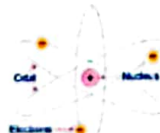


4 Atomic Models

- Plum Pudding Model



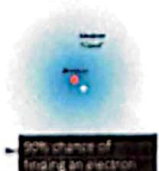
- Rutherford's Model



- Bohr's Model



- Quantum Mechanical



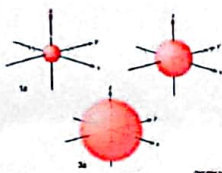
Energy Levels

- Each energy level can be composed of multiple sublevels.
- Energy levels are assigned a number from 1 to 7 based on the row on the periodic table.
- Each sublevel can be composed of multiple orbitals.
- The sublevels are assigned a letter: s, p, d, f, or g.
- Each orbital can hold a maximum of 2 electrons.



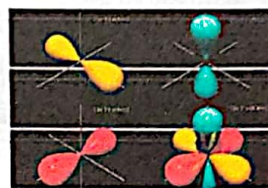
s Sublevel

- All s sublevels have 1 orbital and can hold a maximum of 2 electrons.
- The number in front of s represents the energy level. As the energy level increases, the size of the s sublevel increases, but it can still only hold 2 electrons. (Ex: 1s, 2s, 3s, etc.)
- The s sublevel has a spherical shape.



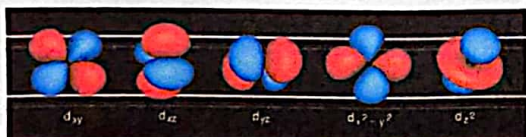
p Sublevel

- All p sublevels have 3 orbitals and can hold a maximum of 6 electrons.
- The p sublevel has a dumbbell or tear drop shape. Each tear drop is referred to as a lobe.



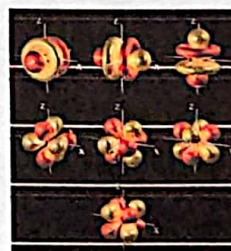
d sublevel

- All d sublevels have 5 orbitals and can hold a maximum of 10 electrons.
- The d sublevel has a four leaf clover shape (4 lobes) or 2 lobes and a donut.



f sublevel

- All f sublevels have 7 orbitals and can hold a maximum of 14 electrons.



g Sublevel

- All g sublevels have 9 orbitals and can hold a maximum of 18 electrons.
- There are not enough elements to fill in the g sublevels yet.



Electrons

Sublevel	# of Orbitals	# of Electrons
s	1	2
p	3	6
d	5	10
f	7	14
g	9	18

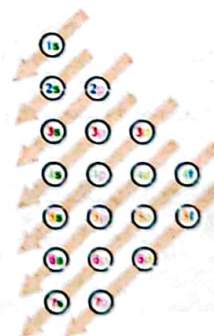
Electron Arrangement in Atoms

- The electron configuration of an atom is the arrangement of the electrons.
- There are 3 rules that govern the electron configuration: Aufbau's principle, Pauli Exclusion principle, and Hund's rule.



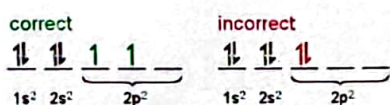
Aufbau's Principle

- Aufbau's principle states that electrons occupy the lowest energy levels first.
- The following is a diagram of the order of the sublevels.



Electron Configuration

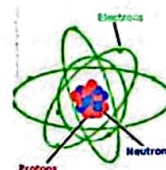
- Pauli Exclusion principle states that an orbital can hold at most 2 electrons.
- When 2 electrons occupy the same orbital, they have opposite spins.
- Hund's rule states that electrons would rather be separate than together in a sublevel with multiple orbitals.



Structure of the Atom

➤ Atom – smallest particle of an element that can exist alone

- Two regions of an atom
 - Nucleus
 - = Center of atom
 - Protons and neutrons
 - Electron "cloud"
 - Area surrounding nucleus containing electrons



Structure of the Atom

- Proton – Positive charge (+), 1 atomic mass unit (amu); found in the nucleus
 - amu - Approximate mass of a proton or a neutron
- Neutron – Neutral charge (0), 1 amu; found in the nucleus
- Electron – Negative charge (-), mass is VERY small!



Counting Atoms

- Atomic Number
 - Number of protons in nucleus
 - The number of protons determines identity of the element!!!

Atomic Number	Atomic Mass	Symbol	Element
5	10.81	B	Boron

6	Atomic number
C	Symbol
12.011	Atomic mass

- Mass Number (Atomic Mass)
 - Number of protons + neutrons
 - Units are g/mol

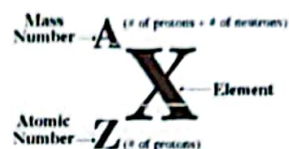
Counting Atoms

➤ Isotopes

- Atoms of the same element with varying number of neutrons
- Different isotopes have different mass numbers because the number of neutrons is different

Counting Atoms

• Nuclear Symbol Notation



Atoms

- Protons have a positive (+) charge and electrons have a negative (-) charge
- In a neutral atom, the number of protons equals the number of electrons, so the overall charge is zero (0)
 - Example/ Helium, with an atomic number of 2, has 2 protons and 2 electrons when stable

Ions

- In a neutral atom
 - Atomic number = # of protons = # of electrons
- Sometimes atoms will gain or lose electrons and form IONS
- Because an electron has a negative charge:
 - When an atom GAINS electrons it becomes NEGATIVE
 - When an atom LOSES electrons it becomes POSITIVE

The Periodic Table of Elements

6 ←

C

CARBON

12 ←

Atomic Number = Number of Protons = Number of Electrons

Chemical Symbol

Chemical Name

Atomic Weight = Number of Protons + Number of Neutrons

KEY

☐ = Solid at room temperature

◐ = Liquid at room temperature

☉ = Gas at room temperature

☼ = Radioactive

⚡ = Artificially Made

57 La Lanthanum	58 Ce Cerium	59 Pr Praseodymium	60 Nd Neodymium	61 Pm Promethium	62 Sm Samarium	63 Eu Europium	64 Gd Gadolinium	65 Tb Terbium	66 Dy Dysprosium	67 Ho Holmium	68 Er Erbium	69 Tm Thulium	70 Yb Ytterbium	71 Lu Lutetium
89 Ac Actinium	90 Th Thorium	91 Pa Protactinium	92 U Uranium	93 Np Neptunium	94 Pu Plutonium	95 Am Americium	96 Cm Curium	97 Bk Berkelium	98 Cf Californium	99 Es Einsteinium	100 Fm Fermium	101 Md Mendelevium	102 No Nobelium	103 Lr Lawrencium

* The atomic weights listed on this Table of Elements have been rounded to the nearest whole number. As a result, this chart actually displays the mass number of a specific isotope for each element. An element's complete, unrounded atomic weight can be found on the IUPAC website: <http://education.jlab.org/elemental/>

The Periodic Table of Elements

LABEL THE COLUMNS!!

6 ←

C

CARBON

12 ←

Atomic Number = Number of Protons = Number of Electrons

Chemical Symbol

Chemical Name

Atomic Weight = Number of Protons + Number of Neutrons*

METALS										NON-METALS							
1 H Hydrogen	2 He Helium																
3 Li Lithium	4 Be Beryllium																
5 B Boron	6 C Carbon	7 N Nitrogen	8 O Oxygen	9 F Fluorine	10 Ne Neon												
11 Na Sodium	12 Mg Magnesium	13 Al Aluminum	14 Si Silicon	15 P Phosphorus	16 S Sulfur	17 Cl Chlorine	18 Ar Argon										
19 K Potassium	20 Ca Calcium	21 Sc Scandium	22 Ti Titanium	23 V Vanadium	24 Cr Chromium	25 Mn Manganese	26 Fe Iron	27 Co Cobalt	28 Ni Nickel	29 Cu Copper	30 Zn Zinc	31 Ga Gallium	32 Ge Germanium	33 As Arsenic	34 Se Selenium	35 Br Bromine	36 Kr Krypton
37 Rb Rubidium	38 Sr Strontium	39 Y Yttrium	40 Zr Zirconium	41 Nb Niobium	42 Mo Molybdenum	43 Tc Technetium	44 Ru Ruthenium	45 Rh Rhodium	46 Pd Palladium	47 Ag Silver	48 Cd Cadmium	49 In Indium	50 Sn Tin	51 Sb Antimony	52 Te Tellurium	53 I Iodine	54 Xe Xenon
55 Cs Cesium	56 Ba Barium	57 La Lanthanum	58 Ce Cerium	59 Pr Praseodymium	60 Nd Neodymium	61 Pm Promethium	62 Sm Samarium	63 Eu Europium	64 Gd Gadolinium	65 Tb Terbium	66 Dy Dysprosium	67 Ho Holmium	68 Er Erbium	69 Tm Thulium	70 Yb Ytterbium	71 Lu Lutetium	
73 Hf Hafnium	74 Ta Tantalum	75 W Tungsten	76 Re Rhenium	77 Os Osmium	78 Ir Iridium	79 Pt Platinum	80 Au Gold	81 Hg Mercury	82 Tl Thallium	83 Pb Lead	84 Bi Bismuth	85 Po Polonium	86 At Astatine	87 Rn Radon			
89 Fr Francium	90 Ra Radium	91 Ac Actinium	92 Th Thorium	93 Pa Protactinium	94 U Uranium	95 Np Neptunium	96 Pu Plutonium	97 Am Americium	98 Cm Curium	99 Bk Berkelium	100 Cf Californium	101 Es Einsteinium	102 Fm Fermium	103 Md Mendelevium	104 No Nobelium	105 Lr Lawrencium	

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I am Dmitri Mendeleev!



Periodic Table

Group numbering is based on the new IUPAC system.
Atomic weights are based on $^{12}\text{C} = 12$ and conform to the 1993 IUPAC report values. Numbers in parentheses are unrounded atomic weights.

1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18
H	He																
Li	Be	B	C	N	O	F	Ne										
Na	Mg	Al	Si	P	S	Cl	Ar										
K	Ca	Sc	Ti	V	Cr	Mn	Fe	Co	Ni	Cu	Zn	Ga	Ge	As	Se	Br	Kr
Rb	Sr	Y	Zr	Nb	Mo	Tc	Ru	Rh	Pd	Ag	Cd	In	Sn	Sb	Te	I	Xe
Cs	Ba		Hf	Ta	W	Re	Os	Ir	Pt	Au	Hg	Tl	Pb	Bi	Po	At	Rn
Fr	Ra		Rf	Db	Sg	Bh	Hs	Mt	Ds	Rg	Uub	Uut	Uuq	Uup	Uuh	Uus	Uuo
Lanthanides		La	Ce	Pr	Nd	Pm	Sm	Eu	Gd	Tb	Dy	Ho	Er	Tm	Yb	Lu	
Actinides		Ac	Th	Pa	U	Np	Pu	Am	Cm	Bk	Cf	Es	Fm	Md	No	Lr	

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I made the PERIODIC TABLE !

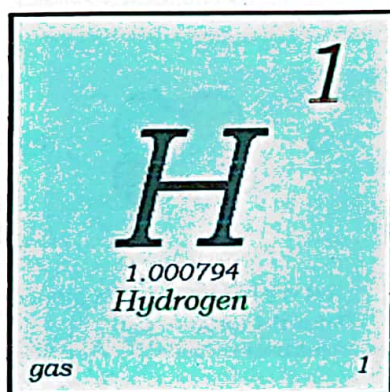
What is the PERIODIC TABLE?

- Shows all known elements in the universe.
- Organizes the elements by chemical properties.

A close-up view of a portion of the periodic table, showing the elements Carbon (C), Nitrogen (N), and Silicon (Si). Each element box includes its atomic number, symbol, name, and atomic weight.

6 C Carbon 12.011	7 N Nitrogen 14.006
14 Si	15

Key to the Periodic Table



- Elements are organized on the table according to their atomic number.

Atomic Number



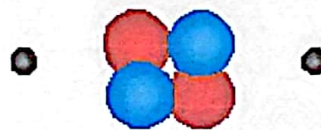
Bohr Model of Hydrogen Atom

- This refers to how many protons an atom of that element has.
- No two elements, have the same number of protons.

Wave Model

Atomic Mass

- Atomic Mass refers to the "weight" of the atom.
- It is derived at by adding the number of protons with the number of neutrons.

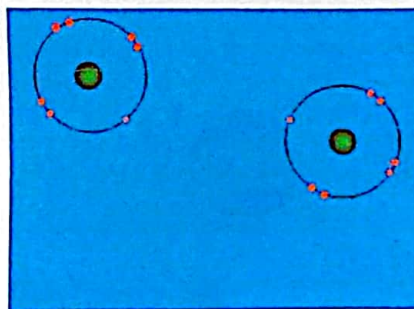


This is a helium atom. Its atomic mass is 4 (protons plus neutrons).

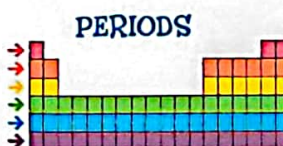
What is its atomic number?

Valence Electrons

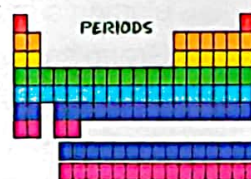
- Valence electrons are the electrons in the outer energy level of an atom.
- These are the electrons that are transferred or shared when atoms bond together.



Rows are called "Periods"



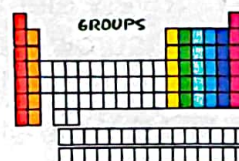
- Periods = rows
- From left to right
- What do elements in a row have in common?
 - the same number of electron shells
- Every element in Period 1 (1st row) has 1 shell for its electrons (H & He)
- All of the elements in period 2 have two shells for their electrons.
- It continues like this all the way down the table



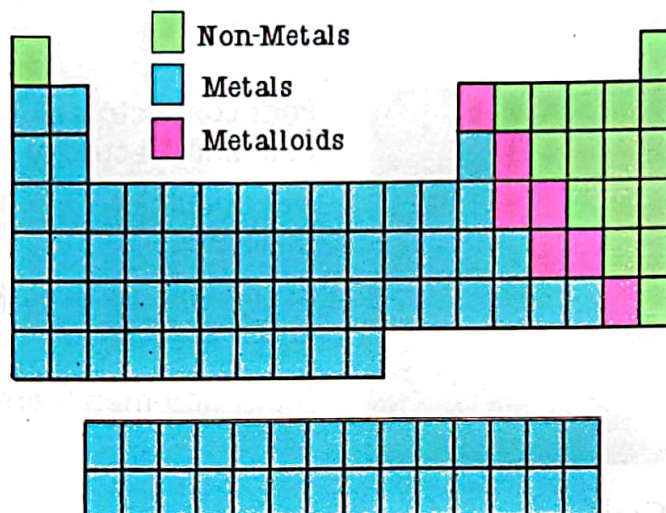
Columns are called "Groups" or Families



- Column = **group** = **families**
- What do elements in a group have in common?
 - same number of valence electrons (electrons in the outer shell)
- They share similar characteristics with the other elements in their family.
- Group 1: 1 valence electron
- Group 2: 2 valence electrons
- Group 13: 3 valence electrons
- Group 14: 4 valence electrons
- Group 15: 5 valence electrons
- Group 16: 6 valence electrons
- Group 17: 7 valence electrons
- Group 18: 8 valence electrons except He who has 2

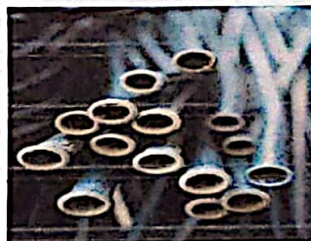


The elements of the periodic table can be divided into three main categories: Metals, Non-Metals, and Metalloids.



Properties of Metals

- Good conductors of heat and electricity
- Shiny.
- Ductile (can be stretched into thin wires)
- Malleable (can be pounded into thin sheets)
- A chemical property of metal is its reaction with water which results in corrosion.



Properties of Non-Metals



- Poor conductors of heat and electricity
- Not ductile or malleable
- Brittle and break easily
- Dull
- Many non-metals are gases.

Sulfur

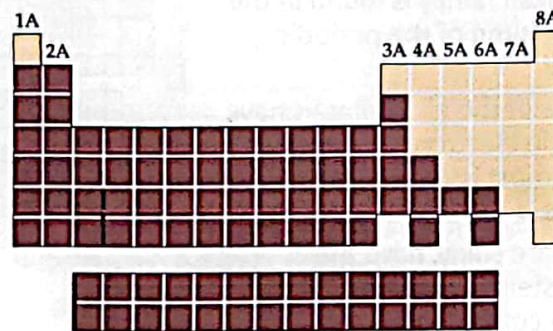
Properties of Metalloids



- Have properties of both metals and non-metals
- Solids that can be shiny or dull.
- Conduct heat and electricity better than non-metals but not as well as metals.
- They are ductile and malleable.

Silicon

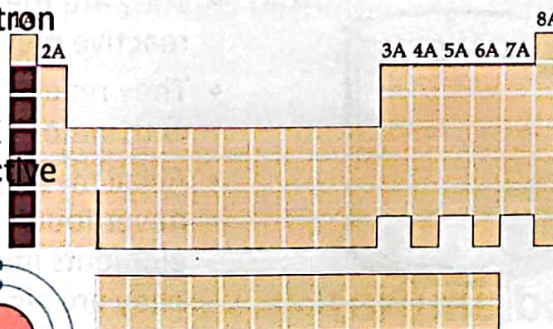
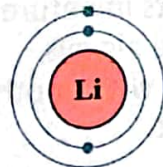
Region: Metals



Group 1: Alkali Metals

- 1 valence electron

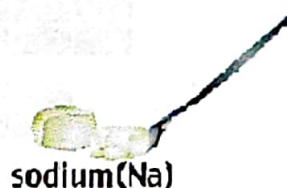
- Very Reactive



Alkali Metals

- The alkali family is found in the first column of the periodic table.
- Atoms of the alkali metals have a single electron in their outermost level, in other words, 1 valence electron.
- They are shiny, have the consistency of clay, and are easily cut with a knife.

Periodic Table of the Elements



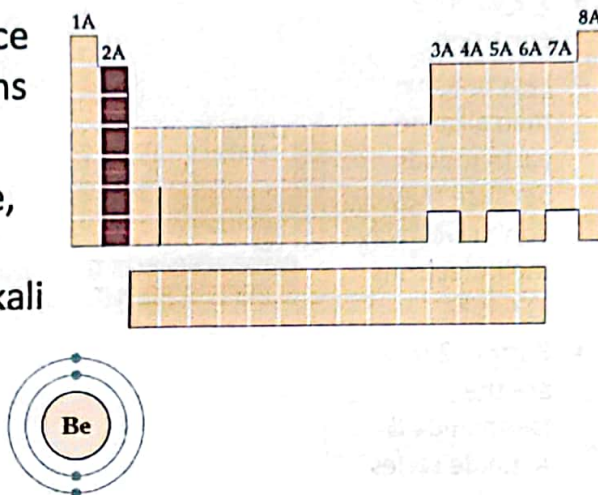
Alkali Metals



- They are the most reactive metals.
- They react violently with water.
- Alkali metals are never found as free elements in nature. They are always bonded with another element.

Group 2: Alkaline Earth Metals

- 2 valence electrons
- Very reactive, but less than alkali metals



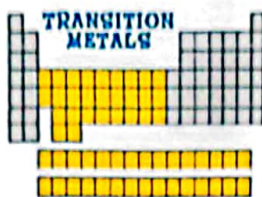
Alkaline Earth Metals

- They are never found uncombined in nature.
- They have two valence electrons.
- Alkaline earth metals include magnesium and calcium, among others.

Periodic Table of the Elements

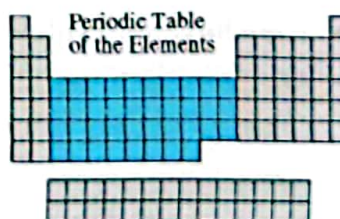
Groups 3 – 12: Transition Metals

- 1-2 valence electrons
- Less reactive than alkaline earth metals because they don't give away their electrons as easily
- Bottom 2 row are the Lanthanide & Actinide series



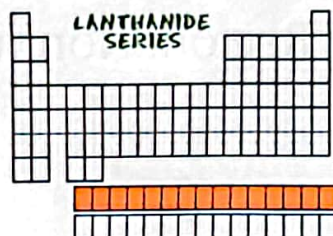
Transition Metals

- Transition Elements include those elements in the B families.
- These are the metals you are probably most familiar: copper, tin, zinc, iron, nickel, gold, and silver.
- They are good conductors of heat and electricity.



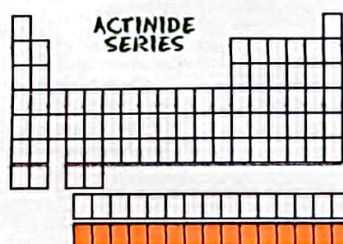
- **Lanthanide Series:**

- shiny reactive metals
- Most found in nature

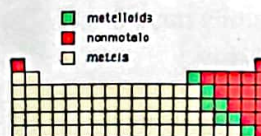


- **Actinides Series:**

- radioactive and unstable
- Most are man-made & not stable in nature



Region: Metalloids



Region: Nonmetals

Boron Family

- ☐ The Boron Family is named after the first element in the family.
- ☐ Atoms in this family have 3 valence electrons.
- ☐ This family includes a metalloid (boron), and the rest are metals.
- ☐ This family includes the most abundant metal in the earth's crust (aluminum).



Periodic Table of the Elements

Carbon Family

- ❑ Atoms of this family have 4 valence electrons.
- ❑ This family includes a non-metal (carbon), metalloids, and metals.
- ❑ The element carbon is called the "basis of life." There is an entire branch of chemistry devoted to carbon compounds called organic chemistry.

Periodic Table of the Elements

Nitrogen Family

- The nitrogen family is named after the element that makes up 78% of our atmosphere.
- This family includes non-metals, metalloids, and metals.
- Atoms in the nitrogen family have 5 valence electrons. They tend to share electrons when they bond.
- Other elements in this family are phosphorus, arsenic, antimony, and bismuth.



Periodic Table of the Elements

Oxygen Family

- Atoms of this family have 6 valence electrons.
- Most elements in this family share electrons when forming compounds.
- Oxygen is the most abundant element in the earth's crust. It is extremely active and combines with almost all elements.

Periodic Table of the Elements

Group 17: Halogens

- 7 valence electrons
- Very reactive
- Nonmetals

Halogen Family

- ❑ The elements in this family are fluorine, chlorine, bromine, iodine, and astatine.

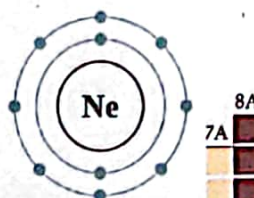
Periodic Table of the Elements

- ❑ Halogens have 7 valence electrons, which explains why they are the most active non-metals. They are never found free in nature.

- ❑ Halogen atoms only need to gain 1 electron to fill their outermost energy level.
- ❑ They react with alkali metals to form salts.

Group 18: Noble Gases

- 8 valence electrons (except He which only has 2)



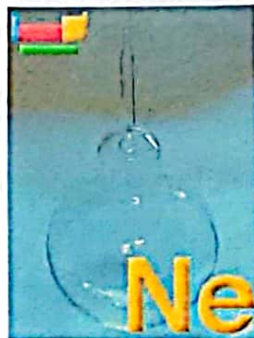
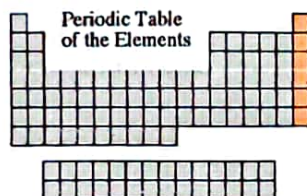
- "Happy" because their outer electron shell is filled!

- NON REACTIVE (inert) gases

- Nonmetals

Noble Gases

Periodic Table of the Elements



- ☐ Noble Gases are colorless gases that are extremely un-reactive.
- ☐ One important property of the noble gases is their inactivity. They are inactive because their outermost energy level is full.
- ☐ Because they do not readily combine with other elements to form compounds, the noble gases are called inert.
- ☐ The family of noble gases includes helium, neon, argon, krypton, xenon, and radon.
- ☐ All the noble gases are found in small amounts in the earth's atmosphere.